

Listening Between the Lines

Analyzing Gendered Conversational Dynamics in Indian Podcasts

Project Documentation — v2 Features and Dyad Analysis

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About This Document

This document records the second version (v2) of the project's linguistic feature analysis of host and guest speech in 418 Indian podcast episodes, in the order the scripts were built. Each section describes the purpose of a script, the methodology it implements, how it was run, and the output it produced.

It covers: `src/compute_bias_features_v2.py` (Section 1, per-speaker hedge, booster and politeness features adapted to speech), the precision audits built with `src/make_audit_sample.py` (Section 2), `src/episode_dyads.py` (Section 3, host→guest gender dyads), `src/dyad_analysis_v2.py` (Section 4, episode-level dyad dataset), `src/stat_analysis/kruskal_wallis_test_v2.py` (Section 5), and `src/stat_analysis/Mann_Whitney_test_v2.py` and `src/stat_analysis/within_host_test_v2.py` (Section 6, pairwise dyad tests and the guest-gender test within host), and `src/speaking_time_v2.py` with `src/stat_analysis/speaking_time_mixed_model_v2.py` (Section 7, host and guest speaking time and a mixed-effects model of the host's share), and the dialogue-act classification of all episodes with `src/stat_analysis/ask_within_host_check_v2.py` (Section 8, a check of the direct-question result with the classifier's question label).

1. `src/compute_bias_features_v2.py`

1.1 Purpose

The project's first feature script counted hedges, boosters, politeness and directives with four hand-built word lists, and a manual check found them unreliable: only 12 of 50 sampled "directive" matches were real directives. This script replaces those lists with two published resources and adapts them to spontaneous podcast speech using repeated precision audits (Section 2). It produces one row of conversational style features per speaker per episode, over all 418 collected episodes.

The two resources are Hyland's (2005) metadiscourse lexicon of hedges and boosters, and the ConvoKit PolitenessStrategies classifier (Danescu-Niculescu-Mizil et al., 2013), which marks 21 politeness strategies (gratitude, apologizing, deference, factuality, direct questions, 1st/2nd person, and others) using a dependency parse of each turn.

1.2 Methodology

- Input: the 418 WhisperX transcripts with diarized speakers, speaker gender, and HOST/GUEST labels (`data/outputs/whisperx/*_whisperx_diarized.gender.host_guest.json`).
- Turns: consecutive segments from the same speaker are merged into one turn before scoring. Segments without a diarized speaker are dropped, and speech in the first and last 60 seconds of each episode (intro/outro monologues and sign-off thank-yous) is excluded.
- Hyland hedges and boosters (`data/lexicons/hyland_2005_hedges_boosters.json`; 101 hedges, 64 boosters): all terms are compiled into one case-insensitive, whole-word regular

expression, longest term first. Each stretch of text is matched at most once, so "no doubt" counts as a booster and does not also count "doubt" as a hedge.

- Spoken-speech adjustments to Hyland, stored in the same file under `spoken_adjustments`: (1) discourse-marker phrases such as "you know", "I don't know", "talk about" and "think about" are matched and discarded, so they cannot count as hedges or boosters; (2) "I think" is counted as a hedge, following Prokofieva & Hirschberg (2014) on hedging in conversation; (3) terms that audited as mostly literal in speech are dropped, e.g. hedges about, around, would, could, should, feel, quite and boosters know, think, thought, find, show, always, never, must, true, believe (with their inflected forms).
- ConvoKit: each episode's turns are loaded into a ConvoKit Corpus, parsed with spaCy (`en_core_web_sm`), and scored with PolitenessStrategies at the marker level, so every occurrence is counted, not just whether a strategy appears in a turn.
- ConvoKit adjustments (`data/lexicons/convokit_adjustments.json`): Factuality no longer counts "really" or "the point"; Deference no longer counts "good"; and Direct_question only counts a wh-question when its sentence ends with "?" and is the last sentence of the turn, so that the other speaker responds next. The reasons and audit counts for each change are recorded in the JSON files and summarized in Section 2.
- For each speaker in each episode, the script writes the word count, turn count, and for every feature the raw count and the rate per 1,000 words. ConvoKit features also get a turn rate: the share of the speaker's turns with at least one occurrence.
- Alongside the features, it writes a long table of which terms or markers produced each count (for tracing any feature back to its words), and a random sample of 50 hits per feature with the surrounding turn text, used for the precision audits.

1.3 How It Was Run

The script runs on the cluster as a CPU-only SLURM job (2 cores, 8 GB, about 21 minutes for 418 episodes) in the convokit conda environment:

```
sbatch sbatch/run_bias_features_v2.sbatch
```

which runs:

```
python src/compute_bias_features_v2.py --input_glob  
"data/outputs/whisperx/*_whisperx_diarized.gender.host_guest.json" --out_csv  
results/features/speaker_features_v2.csv --terms_csv  
results/features/speaker_feature_terms_v2.csv --audit_csv  
results/features/audit_sample_v2.csv --audit_n 50
```

Both sets of adjustments are on by default; `--no_spoken_adjustments` and `--no_convokit_adjustments` run the lexicons as published.

1.4 Output

Output file: `results/features/speaker_features_v2.csv` — 836 rows × 73 columns (one HOST row and one GUEST row for each of the 418 episodes), with no missing values.

File	Rows	Contents
speaker_features_v2.csv	836	Features per (episode, speaker)
speaker_feature_terms_v2.csv	155,750	Count of each term/marker per (episode, speaker, feature)
audit_sample_v2.csv	1150	50 random hits per feature, with turn text

Columns produced:

- episode_id, speaker_id, role, gender, word_count, turn_count.
- hy_hedge_count, hy_hedge_per_1k, hy_booster_count, hy_booster_per_1k (Hyland).
- For each of the 21 ConvoKit strategies: ck_<strategy>_count, ck_<strategy>_per_1k, ck_<strategy>_turn_rate (e.g. ck_factuality_per_1k, ck_direct_question_turn_rate).

1.5 Dataset Composition of the Output

Role	Female	Male	Unknown	Total
HOST	125	293	0	418
GUEST	128	288	2	418

Guests speak far more than hosts: the median guest speaks 8,833 words per episode, against 2,388 for the median host. The two guests with unknown gender are resolved by manual overrides in Section 3.

1.6 Descriptive Statistics

Rates per 1,000 words for the five features carried into the dyad analysis (Sections 4-6), by role, with the words that account for most of each feature's hits:

Feature	Host mean / median	Guest mean / median	Top terms (share of hits)
Hedges	5.55 / 5.00	6.51 / 5.36	i think 41%, maybe 11%, sometimes 8%
Boosters	4.05 / 3.73	4.24 / 3.56	really 32%, actually 22%, of course 10%
Factuality	1.32 / 1.15	1.21 / 0.91	actually 71%, in fact 14%, honestly 7%
Deference	0.26 / 0.00	0.06 / 0.00	interesting 39%, nice 31%, great 22%
Direct questions	4.61 / 3.92	0.29 / 0.17	what 56%, how 26%, why 14%

The spoken-speech exclusions matter: 30,742 phrase matches were discarded ("you know" alone is 61% of them), which is about as many as all counted hedges. Direct questions are almost entirely a host behavior once rhetorical questions are removed. Deference is rare: most speakers never use it, so its median is 0.

1.7 Notes & Limitations

- Word lists cannot read context. The largest remaining source of error is "actually" used as filler ("thank you for actually being here"); it is 71% of Factuality hits and 22% of booster hits. It was kept because it is correct in roughly three quarters of audited uses.
- ConvoKit's strategies were designed on written requests (Wikipedia and Stack Exchange); podcast speech has more fillers, rhetorical questions, and backchannels, which is why the adjustments above were needed.
- Deference in speech is mostly acknowledgement ("Interesting.", "Nice.") rather than explicit praise, so it is better described as acknowledgement.
- Direct_question requires the question to end the turn, which removes about half of the real questions along with the rhetorical ones (Section 2). Rates are comparable across groups but undercount questions in absolute terms. Guest direct questions are not reliable and are not used.
- Diarization errors that merge two speakers into one turn can attribute one person's words to the other.

2. Precision Audits of the v2 Features (src/make_audit_sample.py)

2.1 Purpose

A word list can look plausible and still count mostly the wrong things. Before using the v2 features, each one was checked by hand: a random sample of the words the script counted was read in context, and each hit was marked correct or not. The results drove the adjustments in Section 1.2, and the audit was repeated after each change.

2.2 Methodology

- compute_bias_features_v2.py samples 50 random hits per feature (reservoir sampling) into results/features/audit_sample_v2.csv.
- make_audit_sample.py selects the features being audited, rebuilds each hit's context from the full turn with the matched words in `[[ ]]`, and adds a yes/no question for each feature. For example, for hedges: "Does `[[ ]]` express uncertainty, tentativeness, approximation, or reduced commitment?"; for direct questions: "Is this a direct wh-question actually asked of someone?"
- Each hit is labeled 1 (correct) or 0, with a short note for every 0. Precision is the share of hits labeled 1. Rhetorical and self-answered questions, reported speech, literal uses ("make sure", "the point where"), and filler uses count as 0.
- The labels were made by an AI assistant (Claude) following the per-feature questions, and are marked as such in the labeled_by column of each audit file. They have not yet been checked by a human annotator (see 11.6).

2.3 How It Was Run

```
python src/make_audit_sample.py --features  
hy_hedge,hy_booster,ck_factuality,ck_direct_question,ck_deference --out  
results/audit/audit_v2e_to_label.csv
```

2.4 Audit Rounds

Each round sampled fresh hits from the pipeline as it stood at the time (50 per feature).

Precision per round:

Round	Pipeline	Hedge	Boost.	Fact.	Def.	Q	Change made after
1 (v2)	Published lists + discourse-marker exclusions, "I think" as hedge	0.46	0.34	0.44	0.66	0.44	Drop Hyland terms that were mostly literal (about, would, feel, know, always, thought, ...); drop Factuality "really", Deference "good"; questions must end with "?"
2 (v2b)	After round 1 changes	1.00	0.70	0.84	0.72	0.32	Drop booster "believe" (mostly religious or others' beliefs)
3 (v2c)	After round 2 changes	0.84	0.74	0.76	0.84	0.58	Questions must end the turn; drop Factuality "the point" (0 correct in every round)
4 (v2d)	Final pipeline	—	—	—	—	0.86	None (direct questions only)
5 (v2e)	Final pipeline, fresh sample	0.86	0.68	0.68	0.80	0.86	None

Hedge = hedges, Boost. = boosters, Fact. = factuality, Def. = deference, Q = direct questions.

Round 1 also covered gratitude (0.88) and apologizing (0.80), which were accurate and needed no change; they are not used in the dyad analysis. Round 4's direct-question hits are the same 50 as round 5's.

2.5 Final Precision

For hedges, boosters and deference, rounds 3 and 5 sample the same final lexicon, and factuality only lost "the point", so they are pooled (about 100 hits each; round 3's "the point" hits removed). Direct questions use round 4, the only sample after its last change.

Feature	Hits	Precision (95% CI)	Host	Guest
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Hedges	99	0.86 (0.78-0.91)	0.76 (n=21)	0.88 (n=78)
Boosters	100	0.71 (0.61-0.79)	0.56 (n=27)	0.77 (n=73)
Factuality	95	0.76 (0.66-0.83)	0.65 (n=20)	0.79 (n=75)
Deference	98	0.82 (0.73-0.88)	0.92 (n=53)	0.69 (n=45)
Direct questions	50	0.86 (0.74-0.93)	0.97 (n=40)	0.40 (n=10)

All five features are between 0.71 and 0.86 overall. Two findings shaped the dyad analysis: guest direct questions stayed unreliable (mostly rhetorical questions inside stories), so direct questions are analyzed for hosts only; and booster and deference precision differs between hosts and guests, so the analysis compares each role separately across dyads rather than subtracting guest rates from host rates (Section 4.2). In the audit samples, precision for female and male speakers was similar, but there were only 5-16 female speakers per feature.

2.6 Notes & Limitations

- With 50 hits, a single round's precision is uncertain by about ± 0.10 ; changes between rounds smaller than that are within noise.
- The rule that questions must end the turn was chosen using the round 3 labels, which overstates how well it works on those labels (0.88). Round 4 checks it on new hits (0.86).
- Labels were made by an AI assistant and one pass only; a human check of a subset (for example 50 hits per feature, with inter-annotator agreement) would make these numbers citable.
- The audit snippet marks the first place the term appears in the turn, which is not always the hit that was counted (e.g. a turn with several questions). Affected rows were labeled from the real hit and noted.

3. src/episode_dyads.py

3.1 Purpose

Assigns every episode its host→guest gender dyad, using all 418 episodes rather than a balanced subsample, so no episodes are discarded to equalize group sizes.

3.2 Methodology

- From speaker_features_v2.csv, take the gender of the HOST and of the GUEST speaker with the most words in each episode.
- Apply manual guest-gender corrections for the two episodes where gender detection returned "unknown" (ep_174 and ep_261, both male after review).
- Keep episodes whose dyad is one of MALE→MALE, MALE→FEMALE, FEMALE→MALE, FEMALE→FEMALE (host gender first); none had to be dropped.

3.3 How It Was Run

```
python src/episode_dyads.py --features results/features/speaker_features_v2.csv --out results/features/episode_dyads.csv
```

3.4 Output

Output file: results/features/episode_dyads.csv — 418 rows (episode_id, host_gender, guest_gender, dyad).

Dyad	Episodes
MALE->MALE	226
MALE->FEMALE	67
FEMALE->MALE	64
FEMALE->FEMALE	61
Total	418

3.5 Who the Hosts Are

The corpus has only 9 hosts, identified by matching the host's voice to the speaker library (data/speaker_library.json). Every host has one gender, so all female-host episodes come from 4 hosts and all male-host episodes from 5:

Host	MM	MF	FM	FF	Total
Faye Dsouza	0	0	19	14	33
Karishma Mehta	0	0	28	32	60
Pooja Dhingra	0	0	4	5	9
Soha Ali Khan	0	0	1	1	2
Nikhil Kamath	15	0	0	0	15
Prakhar Gupta	18	5	0	0	23
Raj Shamani	117	32	0	0	149
Ranveer A	49	26	0	0	75
Samdish Bhatia	19	4	0	0	23
Not matched	8	0	12	9	29

MM = MALE->MALE, MF = MALE->FEMALE, FM = FEMALE->MALE, FF = FEMALE->FEMALE. One host, Raj Shamani, hosts 149 episodes, including 117 of the 226 MALE->MALE ones. In 29 episodes the host was labeled by the fallback method rather than a library match, so their identity is not recorded.

3.6 Notes & Limitations

- The dyads are unbalanced (226 MALE->MALE against 61-67 in each other group). The tests in Sections 5-6 are rank-based and do not assume equal group sizes.

- Because host gender is tied to 9 individual hosts and their shows, any difference between male-host and female-host dyads may be about those particular hosts rather than gender. Section 6 adds a test that holds the host fixed.

4. src/dyad_analysis_v2.py

4.1 Purpose

Builds the episode-level dyad dataset from the v2 features, with descriptive summaries by dyad and by host. The statistical tests on this dataset are separate scripts in src/stat_analysis/ (Sections 5 and 6).

4.2 Methodology

- Join speaker_features_v2.csv to episode_dyads.csv, and put each episode's host and guest values side by side (one row per episode).
- Metrics, all per 1,000 words: hedges, boosters, factuality and deference for both host and guest; direct questions for the host only (Section 2.5). Plus host speaking share (host words / host + guest words) and dominance ratio (host words / guest words).
- Host and guest are compared separately, not as a host-minus-guest difference. Because booster and deference precision differs between hosts and guests (Section 2.5), subtracting one from the other would mix two different error rates. Each role is instead compared across dyads, e.g. host hedging with male guests vs. female guests.
- Host identity (host_name) is read from the speaker-library match in each transcript.

4.3 How It Was Run

```
python src/dyad_analysis_v2.py
```

All paths default to the files above.

4.4 Output Files

File	Contents
results/dyads/dyad_analysis_v2.csv	418 rows: dyad, host name, host/guest metrics
results/dyads/dyad_median_mad_stats_v2.csv	Median, MAD, n per dyad per metric
results/dyads/host_level_medians_v2.csv	Median of each metric for each of the 9 hosts

4.5 Medians by Dyad

Metric	MM	MF	FM	FF
Host speaking share	0.22	0.28	0.18	0.20
Host hedges	5.04	5.46	5.04	4.64
Guest hedges	4.37	4.86	8.21	7.13
Host boosters	3.05	3.51	5.28	5.98

Guest boosters	2.47	3.46	4.70	6.11
Host factuality	0.96	1.19	1.55	1.28
Guest factuality	0.74	0.96	1.03	1.75
Host deference	0.00	0.24	0.00	0.00
Guest deference	0.00	0.00	0.00	0.00
Host direct questions	3.33	5.15	6.36	3.70

Deference medians are 0 in every dyad because most speakers never use it; the share of episodes where the host uses it at least once is MM 46%, MF 60%, FM 5%, FF 10%.

4.6 Medians by Host

Host	Sex	Eps.	Host share	Host hedges	Host boosters	Host fact.	Host dir. Q	Guest boosters
Faye Dsouza	F	33	0.19	4.12	5.85	1.83	4.15	4.59
Karishma Mehta	F	60	0.18	4.67	5.82	1.09	6.62	5.13
Pooja Dhingra	F	9	0.24	7.53	6.24	2.07	3.43	7.51
Soha Ali Khan	F	2	0.22	8.79	4.24	0.86	2.36	3.42
Nikhil Kamath	M	15	0.22	10.93	3.59	1.03	3.45	6.43
Prakhar Gupta	M	23	0.30	5.31	3.79	1.52	2.38	2.76
Raj Shamani	M	149	0.22	4.14	2.28	0.85	3.88	2.23
Ranveer A	M	75	0.25	7.81	4.04	1.43	3.79	3.17
Samdish Bhatia	M	23	0.22	4.77	3.26	0.47	4.96	2.79

Hosts of the same gender differ as much as the two genders do. For example, host hedging ranges from 4.1 to 10.9 per 1,000 words among male hosts, and from 4.1 to 8.8 among female hosts. This is why the host-gender comparisons in Sections 5-6 need to be read as comparisons between these particular hosts.

5. src/stat_analysis/kruskal_wallis_test_v2.py

5.1 Purpose

Tests whether each metric in results/dyads/dyad_analysis_v2.csv (Section 4) differs across the four gender dyads at all, before asking which dyads differ (Section 6).

5.2 Methodology

- For each of the 11 metrics in Section 4.2, run scipy's `kruskal()` across the four dyad groups (non-parametric, rank-based, no normality assumption).
- Correct the 11 p-values for multiple testing with the Benjamini-Hochberg false discovery rate (FDR), and call a metric significant if its corrected p-value is below 0.05.

- Save H, raw p, corrected p, and the median of each dyad to results/stat_analysis/kruskal_wallis_v2_results.csv.

5.3 How It Was Run

python src/stat_analysis/kruskal_wallis_test_v2.py

Paths default to the files above. The Benjamini-Hochberg correction is in src/stat_analysis/fdr.py, shared with the Section 6 scripts (statsmodels is not installed in the convokit environment).

5.4 Results

Metric	H statistic	p-value	FDR p	Significant
Host speaking share	50.50	<0.0001	<0.0001	Yes
Dominance ratio	50.50	<0.0001	<0.0001	Yes
Host hedges	3.26	0.3528	0.3528	No
Guest hedges	50.44	<0.0001	<0.0001	Yes
Host boosters	96.56	<0.0001	<0.0001	Yes
Guest boosters	74.27	<0.0001	<0.0001	Yes
Host factuality	12.68	0.0054	0.0059	Yes
Guest factuality	50.76	<0.0001	<0.0001	Yes
Host deference	65.54	<0.0001	<0.0001	Yes
Guest deference	20.25	0.0002	0.0002	Yes
Host direct questions	34.74	<0.0001	<0.0001	Yes

10 of the 11 metrics differ significantly across dyads. Only host hedging does not ($p = 0.35$): hosts hedge at similar rates in all four dyads.

5.5 Notes & Limitations

- Host speaking share and dominance ratio are two transformations of the same word counts, so their rank-based results are identical; they are one finding, not two.
- A significant result means at least one dyad differs. Because host gender is tied to 9 specific hosts (Section 3.5), a significant host-side metric can reflect differences between those hosts rather than gender. Section 6 separates the two for guest gender.

6. src/stat_analysis/Mann_Whitney_test_v2.py and within_host_test_v2.py

6.1 Purpose

Identifies which pairs of dyads differ for each significant metric, and then checks which differences survive when the host is held fixed. Since each host interviews both female and male guests, guest-gender effects can be tested within each host's show; host-gender effects cannot.

6.2 Methodology

- Pairwise (Mann_Whitney_test_v2.py): for each of the 10 metrics significant in Section 5 (read from kruskal_wallis_v2_results.csv), run a two-sided Mann-Whitney U test on all 6 dyad pairs, correct the 6 p-values with Benjamini-Hochberg FDR within each metric, and compute the rank-biserial correlation as effect size. A positive value means dyad A is higher.
- Within host (within_host_test_v2.py): episodes are ranked within each host (as a percentile, 0-1) on each metric. For each host, compute the mean rank of female-guest episodes minus the mean rank of male-guest episodes, then average across hosts, weighting hosts with more of both guest genders more heavily. A value of +0.15 means female-guest episodes sit, on average, 15 percentile points higher within their host's episodes.
- The p-value comes from a permutation test: guest gender is shuffled within each host 10,000 times, and p is the share of shuffles with an effect at least as large. P-values are FDR-corrected across the 10 metrics. Episodes without a library-matched host are left out (8 hosts with both guest genders, 374 episodes).
- Exploratory: the same within-host effect is computed separately for male and female hosts, and their difference is tested with the same shuffles (does the guest-gender effect depend on the host's gender?).

6.3 How It Was Run

```
python src/stat_analysis/Mann_Whitney_test_v2.py
```

```
python src/stat_analysis/within_host_test_v2.py
```

Mann_Whitney_test_v2.py must run after kruskal_wallis_test_v2.py, since it tests only the metrics that were significant there. within_host_test_v2.py takes --n_perm (default 10,000) and --seed (default 42) for the permutation test. Outputs:

results/stat_analysis/pairwise_dyad_tests_v2.csv and
results/stat_analysis/within_host_guest_gender_v2.csv.

6.4 Results: Pairwise Differences Between Dyads

Of the 54 pairwise tests (9 metrics × 6 pairs, leaving out dominance ratio, which repeats speaking share), 38 are significant after FDR correction. Each cell shows the rank-biserial effect size for dyad A vs. dyad B (positive = A higher); * marks a significant pair.

Metric	MM vs MF	MM vs FM	MM vs FF	MF vs FM	MF vs FF	FM vs FF
Host speaking share	-0.37*	+0.32*	+0.18*	+0.65*	+0.54*	-0.16
Guest hedges	-0.15	-0.48*	-0.40*	-0.42*	-0.32*	+0.07
Host boosters	-0.17*	-0.50*	-0.72*	-0.39*	-0.62*	-0.19
Guest boosters	-0.23*	-0.44*	-0.62*	-0.28*	-0.49*	-0.23*
Host factuality	-0.07	-0.24*	-0.21*	-0.18	-0.14	+0.04
Guest factuality	-0.18*	-0.28*	-0.57*	-0.13	-0.45*	-0.23*
Host deference	-0.14	+0.40*	+0.38*	+0.53*	+0.51*	-0.05
Guest deference	-0.01	+0.25*	+0.22*	+0.26*	+0.24*	-0.03

Host direct questions	-0.27*	-0.44*	-0.12	-0.21	+0.16	+0.33*
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The biggest differences separate male-host from female-host dyads (MM and MF vs. FM and FF): boosters are higher in female-host episodes for both hosts and guests, factuality is higher in female-host episodes than in MALE->MALE ones, and host deference is higher in male-host episodes. Host speaking share is highest in MF episodes (median 0.28, vs. 0.18-0.22 elsewhere). Host direct questions are highest in FM episodes.

6.5 Results: Guest Gender Within Host

Metric	Effect (female – male guests)	p (perm.)	FDR p	Significant	Hosts in same direction
Host speaking share	+0.149	<0.0001	0.0003	Yes	6 of 7
Host hedges	+0.013	0.6898	0.8623	No	5 of 7
Guest hedges	+0.055	0.1010	0.1683	No	5 of 7
Host boosters	+0.085	0.0118	0.0236	Yes	6 of 7
Guest boosters	+0.154	<0.0001	0.0003	Yes	7 of 7
Host factuality	-0.001	0.9871	0.9871	No	5 of 7
Guest factuality	+0.131	<0.0001	0.0003	Yes	6 of 7
Host deference	+0.067	0.0111	0.0236	Yes	5 of 7
Guest deference	+0.004	0.9044	0.9871	No	3 of 7
Host direct questions	+0.025	0.4584	0.6548	No	3 of 7

"Hosts in same direction" counts the hosts whose female-guest episodes rank higher, out of the 7 hosts with at least 4 episodes of each guest gender (Soha Ali Khan, with one of each, is left out of the count but not the test).

6.6 Results: Does the Guest-Gender Effect Depend on Host Gender? (Exploratory)

Metric	Effect, male hosts	Effect, female hosts	p (perm.)	FDR p
Host speaking share	+0.169	+0.112	0.4196	0.9332
Host hedges	+0.028	-0.015	0.5464	0.9332
Guest hedges	+0.106	-0.043	0.0359	0.1795
Host boosters	+0.093	+0.070	0.7499	0.9332
Guest boosters	+0.151	+0.157	0.9332	0.9332
Host factuality	+0.002	-0.005	0.9299	0.9332
Guest factuality	+0.120	+0.152	0.6605	0.9332
Host deference	+0.092	+0.021	0.1283	0.4276
Guest deference	-0.003	+0.016	0.7240	0.9332
Host direct questions	+0.145	-0.205	<0.0001	0.0010

Only host direct questions differ after correction: male hosts ask female guests more direct questions than male guests, while female hosts ask female guests fewer. This test was added after the per-host results showed the pattern, so Section 8 checks it with an independent measure of questions.

6.7 Interpretation: What These Results Suggest

- **Hosts take more of the floor with female guests.** Holds within host: hosts talk more when the guest is a woman (+0.15, in 6 of 7 hosts). The pooled result (MF highest) is therefore not just a difference between shows.
- **Female guests speak with more emphasis and certainty, not more hedging.** Within the same show, female guests use more boosters (+0.15, all 7 hosts) and more factuality markers (+0.13) than male guests. They do not hedge more: the pooled difference in guest hedging (higher in female-host dyads) disappears within host, so it came from the shows, not the guests' gender.
- **Hosts adjust slightly to female guests.** Within host, hosts use more boosters (+0.09) and more acknowledgements (+0.07) with female guests. Host hedging and host factuality do not change with guest gender.
- **Host-gender differences cannot be separated from who the host is.** Female-host episodes have more boosters and factuality, and male-host episodes more deference, but these are differences between 4 and 5 individual hosts, and hosts of the same gender vary as much as the genders do (Section 4.6). They cannot be attributed to host gender with this corpus.
- **Direct questions may follow a cross-gender pattern.** Pooled, FM episodes have the most host direct questions but FF episodes do not, so this is not simply a host-gender difference. The exploratory within-host result suggests hosts ask more direct questions of guests of the other gender; Section 8 confirms it with the dialogue-act classifier's question label.

6.8 Notes & Limitations

- The within-host test answers a narrower question than the pooled test: whether the same host behaves differently with female and male guests. It cannot test host gender, because no host appears in both host-gender groups.
- These are associations in observational data. Guests are not randomly assigned to hosts, so female and male guests on the same show may also differ in profession or topic, which this analysis does not control for and which could explain part of the guest-gender effects.
- Feature precision is 0.71-0.86 (Section 2). Errors that are the same across dyads weaken differences rather than create them; errors that differ by guest gender could not be ruled out with the small number of female speakers in the audits.
- FDR correction is applied within each test family (11 omnibus metrics; 6 pairs per metric; 10 within-host metrics), not across all tests together.

7. src/speaking_time_v2.py and src/stat_analysis/speaking_time_mixed_model_v2.py

7.1 Purpose

Measures how much of each conversation the host and the guest speak, in time rather than words, and tests whether the host's share of speaking time depends on the host->guest gender dyad (MM, MF, FM, FF), using a linear mixed-effects model that separates gender effects from show-to-show variation.

7.2 Methodology

- Speaking time: for each speaker in each episode, the sum of (end - start) over their diarized WhisperX segments. As in Section 1, segments without a speaker are dropped and segments starting in the first or last 60 seconds of the episode are excluded. The host and guest of an episode are the HOST and GUEST speakers with the most words (Section 3).
- Outcome: the host's share of the host-plus-guest speaking time, one value per episode. The guest's share is its complement, so modeling both speakers would add no information.
- Model: $\text{host_time_share} \sim \text{host_female} * \text{guest_female} + (1 | \text{host})$. The two gender indicators and their interaction span the four dyads. The random intercept is per host (show), the level at which show-to-show variation lives; an episode-level random intercept is not identifiable, since each episode contributes one value.
- The model uses the 389 episodes whose host matched the speaker library (9 hosts). The 29 episodes without a matched host have no host label and are excluded (MM 8, MF 0, FM 12, FF 9).
- Estimation: restricted maximum likelihood (statsmodels MixedLM). With only 9 hosts, p-values use between-within degrees of freedom, a small-sample correction: terms that vary only between hosts (intercept, host gender) get 7 df (hosts minus 2); terms that vary within hosts (guest gender, interaction) get 378.
- Reported: fixed effects; the model-estimated host share per dyad with 95% CIs; the 6 pairwise dyad contrasts (Holm-corrected); variance components, ICC and R² (Nakagawa's marginal and conditional); likelihood-ratio tests of the host random effect (boundary-corrected) and of the interaction.
- Checks: the outcome on the logit scale; word share instead of time share; the balanced 200-episode subset; each model re-fit leaving one host out; an OLS model with host fixed effects (guest-gender effect estimated strictly within host); and the rank-based within-host permutation test of Section 6.

7.3 How It Was Run

```
python src/speaking_time_v2.py
```

```
python src/stat_analysis/speaking_time_mixed_model_v2.py
```

Both run in the convokit environment (the model needs statsmodels). Paths default to the files below.

7.4 Output Files

File	Contents
results/features/speaking_time_v2.csv	Speaking time per (episode, speaker)
results/dyads/speaking_time_dyads_v2.csv	One row per episode: dyad, host, minutes, host share
results/dyads/speaking_time_by_dyad_v2.csv	Descriptives per dyad
results/dyads/speaking_time_by_host_v2.csv	Median host share per host and guest gender
results/stat_analysis/speaking_time_mixed_model_v2_*.csv	Fixed effects, estimated means, contrasts, variance, robustness, leave-one-host-out, rank test
results/figures/speaking_time_share_by_dyad.png	Figure 7.1
results/figures/speaking_time_model_diagnostics.png	Figure 7.2

7.5 Speaking Time by Dyad

Dyad	Episodes	Median host share	IQR	Host min	Guest min	Host+guest min
MALE->MALE	226	21.3%	17.0%-28.1%	13.8	49.1	64.3
MALE->FEMALE	67	26.3%	22.6%-32.8%	18.0	49.0	69.2
FEMALE->MALE	64	19.0%	15.5%-21.4%	7.8	35.9	43.3
FEMALE->FEMALE	61	19.8%	15.2%-25.0%	8.0	36.3	45.6

All 418 episodes; minutes are medians per episode, after excluding the first and last minute. Guests speak for most of every conversation: the median host share is 19-26% across dyads. Episodes hosted by men are longer (about 65-70 minutes of host and guest speech, against about 45 for episodes hosted by women). Time-based and word-based host share agree closely ($r = 0.986$).

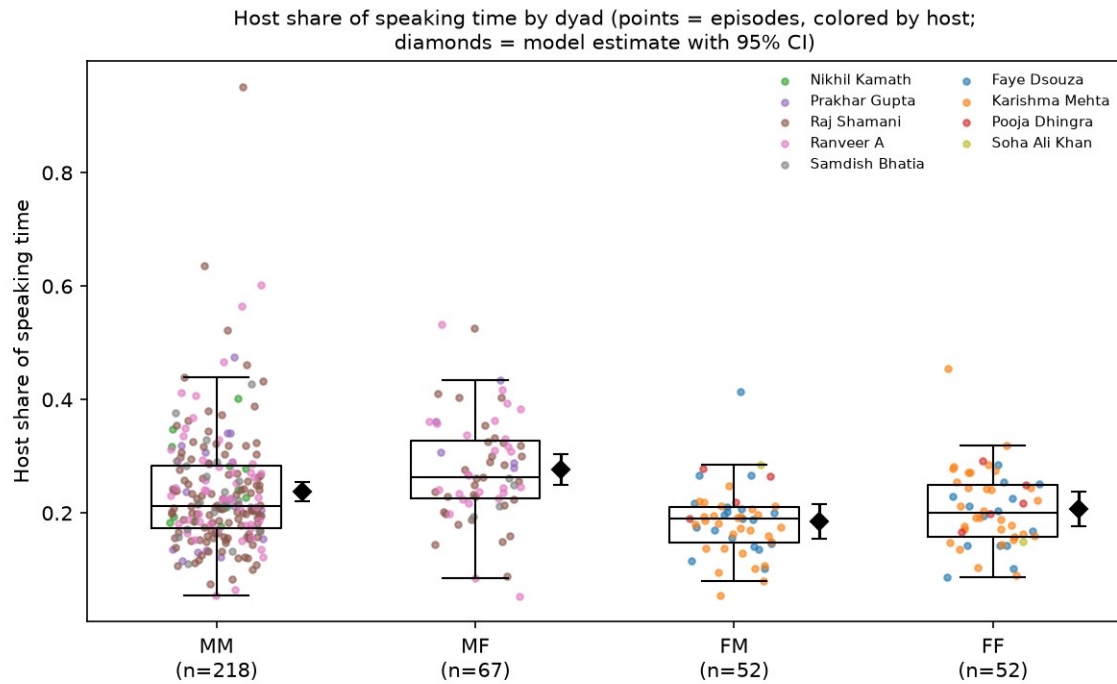


Figure 7.1. Host share of speaking time by dyad for the episodes in the model. Points are episodes, colored by host; boxes show the median and IQR; diamonds are the model estimates with 95% CIs.

Median host share for each host, by guest gender:

Host	Sex	Eps. (F guest)	Eps. (M guest)	Share, F guest	Share, M guest	Difference
Faye Dsouza	F	14	19	19.9%	19.0%	+0.9 pts
Karishma Mehta	F	32	28	19.8%	17.7%	+2.0 pts
Pooja Dhingra	F	5	4	21.7%	24.2%	-2.5 pts
Soha Ali Khan	F	1	1	15.0%	28.5%	-13.6 pts
Nikhil Kamath	M	—	15	—	21.1%	—
Prakhar Gupta	M	5	18	30.7%	21.9%	+8.7 pts
Raj Shamani	M	32	117	25.9%	20.8%	+5.1 pts
Ranveer A	M	26	49	26.7%	23.1%	+3.6 pts
Samdish Bhatia	M	4	19	23.1%	22.3%	+0.8 pts

Six of the seven hosts with several episodes of each guest gender take a larger share with female guests (Pooja Dhingra, with 5 and 4 episodes, is the exception; Soha Ali Khan has one of each).

7.6 Model Results

Term	Estimate	SE	df	95% CI	p
------	----------	----	----	--------	---

Intercept (male host, male guest)	+0.2375	0.0070	7	+0.221 to +0.254	<0.0001
Host female	-0.0526	0.0145	7	-0.087 to -0.018	0.0083
Guest female	+0.0393	0.0131	378	+0.014 to +0.065	0.0029
Host female × guest female	-0.0176	0.0225	378	-0.062 to +0.027	0.4342

Estimates are in share units (0.039 = 3.9 percentage points). Model-estimated host share per dyad:

Dyad	Episodes	Estimated host share	95% CI
MALE->MALE	218	23.7%	22.1%-25.4%
MALE->FEMALE	67	27.7%	25.0%-30.4%
FEMALE->MALE	52	18.5%	15.4%-21.6%
FEMALE->FEMALE	52	20.7%	17.6%-23.7%

Pairwise contrasts (difference in host share, first minus second; Holm-corrected across the 6):

Contrast	Difference	Type	df	p	p (Holm)
MM - MF	-3.9 pts	within host (guest gender)	378	0.0029	0.0145
MM - FM	+5.3 pts	between hosts	7	0.0083	0.0250
MM - FF	+3.1 pts	between hosts	7	0.0700	0.1400
MF - FM	+9.2 pts	between hosts	7	0.0011	0.0065
MF - FF	+7.0 pts	between hosts	7	0.0047	0.0188
FM - FF	-2.2 pts	within host (guest gender)	378	0.2357	0.2357

7.7 Variance Components and Model Comparisons

Quantity	Value
Between-host (show) variance	1.85e-11
Residual (episode) variance	0.0087
ICC (share of variance due to host)	0.000
R2 marginal (fixed effects)	0.079
R2 conditional (fixed effects + host)	0.079
LRT, host random effect (p, boundary-corrected)	0.50
LRT, host × guest gender interaction (chi2, p)	0.62, 0.43

The between-host variance is estimated at zero (a singular fit): once host gender is in the model, the shows do not differ in host share beyond episode-to-episode noise, and the host random effect does not improve the fit. Host and guest gender together explain about 8% of the variation in host share (R2 marginal); the remaining variation is between episodes. With zero host variance the mixed model is equivalent to an ordinary regression and no longer adjusts for which host an episode belongs to, so the guest-gender result is also checked within host below.

7.8 Robustness

Model	Eps.	Guest female: mixed model	Guest female: host fixed effects	Host female	Interaction p
Main model (time share)	389	+3.9 pts (p 0.0029)	+3.7 pts (p 0.0055)	-5.3 pts (p 0.0083)	0.4342
Logit of time share	389	+0.213 (p 0.0049)	+0.207 (p 0.0070)	-0.305 (p 0.0079)	0.5770
Word share	389	+4.5 pts (p 0.0007)	+4.3 pts (p 0.0012)	-5.9 pts (p 0.0050)	0.3840
Balanced 200-episode subset	181	+1.5 pts (p 0.4598)	+1.4 pts (p 0.5139)	-7.6 pts (p 0.0246)	0.8762

Guest-female effects are for male hosts (the reference); the logit row is on the logit scale. The host-fixed-effects column estimates the guest-gender effect strictly within each host.

Rank-based within-host test (Section 6 method) on host time share: +0.126 (p 0.0002) on all episodes and +0.099 (p 0.0236) on the balanced subset.

Leaving out one host at a time, the guest-female effect stays between +3.4 pts and +4.2 pts with p at most 0.014. The host-female effect stays negative but loses significance when Karishma Mehta (p 0.19) or Ranveer A (p 0.55) is left out. (The fit without Ranveer A does not formally converge because the host variance is on the zero boundary; the highest-likelihood fit is reported.)

In the balanced subset the mean-based estimates of the guest effect are small (+1.4 to +1.5 pts, not significant), while the rank-based test is significant. The difference comes from a few male-guest episodes of Raj Shamani with unusually high host share that are in the balanced sample (mean 26.0% against 22.1% for his other male-guest episodes, with the same median of 21%); means are sensitive to them, ranks are not.

7.9 Diagnostics

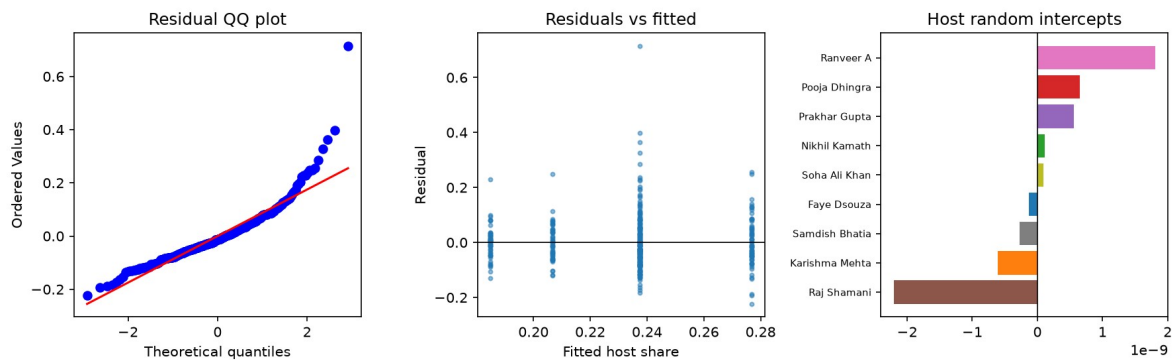


Figure 7.2. Residual QQ plot, residuals against fitted values, and the host random intercepts (all near zero, on a 10^{-9} scale).

Residuals are right-skewed: a few episodes have a very high host share. The largest, ep_202 (Raj Shamani, MALE->MALE), gives the host 95.2% of speaking time and the guest 1.2 minutes, most likely a diarization error. The logit-scale fit and the rank-based test, which are less sensitive to such values, give the same conclusions.

7.10 Interpretation: What These Results Suggest

- **Hosts take more of the conversation with female guests.** With male hosts, the host's share of speaking time is +3.9 pts higher when the guest is a woman (27.7% vs. 23.7%; $p = 0.003$). The effect is the same within host (+3.7 pts), holds without any single host, on the logit scale, with word share, and in the rank-based test, including the balanced subset.
- **The effect for female hosts is smaller and uncertain.** Female hosts also take a larger share with female guests (20.7% vs. 18.5%), but this difference is not significant ($p = 0.24$), and the interaction is not significant either ($p = 0.43$). The data are consistent with the same guest-gender effect for both host genders, with less precision for the 4 female hosts' 104 episodes.
- **The host-gender difference describes these particular shows.** The model estimates that female hosts take 5.3 pts less of the conversation than male hosts ($p = 0.008$ with 7 df), but this is a comparison of 4 female-hosted and 5 male-hosted shows, and it loses significance when Karishma Mehta or Ranveer A is left out. It describes these shows; it cannot be attributed to host gender.
- **There is little show-level variation beyond host gender.** The between-host variance is zero once host gender is included: apart from the host-gender difference, the shows do not differ systematically in how much the host speaks, and most of the variation is between episodes of the same show.

7.11 Notes & Limitations

- Speaking time comes from diarization. Overlapping speech is counted for both speakers, and misattributed segments move time from one speaker to the other; episodes such as the one in 7.9 are likely diarization errors.
- With 9 hosts, all host-level quantities (host-gender effect, between-host variance) are estimated from very little information; the between-within degrees of freedom make the tests conservative, but they cannot add information that is not there.
- The model assumes normally distributed residuals; they are right-skewed, which is why the logit and rank-based checks are reported.
- These are associations in observational data. Guests are not randomly assigned to shows, and female and male guests may differ in profession or topic, which could account for part of the guest-gender difference.

8. Dialogue Acts for All Episodes and a Check of the Direct-Question Result

8.1 Purpose

Section 6 found an exploratory pattern in host direct questions: with the host held fixed, male hosts ask female guests more direct questions than male guests, while female hosts ask female guests fewer. Because the test was chosen after the per-host results suggested the pattern, it needs confirmation with a second, independent way of measuring questions. This section classifies every host and guest segment of all 418 episodes into dialogue acts with a pretrained model, and repeats the within-host test on the model's "ask" label.

8.2 Dialogue-Act Classification (`src/dialogue_act_classification.py`)

- Model: diwank/silicone-deberta-pair, a DeBERTa model fine-tuned on the SILICONE dialogue-act benchmark, which assigns one of 11 labels: ask, answer, say, acknowledge, intent, reply_yes, reply_no, backchannel, exclaim, hold, ask_yes_no.
- Unit: each WhisperX segment with text, classified together with the preceding segment as context. Unlike Sections 1-7, segments are not merged into turns and the first and last 60 seconds are not excluded; the label files carry no timestamps.
- Runs: the script classifies the episodes listed in a CSV. It was run once for 200 episodes (4 h 36 min on one A100 GPU) and once, with the same script and settings, for the remaining 218 (5 h 19 min); `src/combine_dialogue_acts.py` joins the two into 418-episode tables and checks that no episode appears twice.

The 418 episodes give 556,045 segments (143,207 host, 412,838 guest). Share of segments by label (labels under 0.5% for both roles omitted):

Label	Host segments	Guest segments
answer	36.9%	42.0%
say	29.6%	42.3%
ask	19.9%	7.5%
acknowledge	4.9%	2.4%
intent	3.7%	2.7%
reply_yes	2.7%	1.8%
backchannel	1.4%	0.8%
ask_yes_no	0.3%	0.2%

"ask" is the model's general question label: it is applied to wh-questions and most yes/no questions alike (ask_yes_no is rare). One in five host segments is a question, against one in thirteen guest segments.

8.3 Methodology of the Check (src/stat_analysis/ask_within_host_check_v2.py)

- Host question measures per episode, from all segments labeled HOST: the number of “ask” segments per 1,000 host words (comparable to the direct-question rate, which is also per 1,000 words), and the share of host segments labeled “ask”.
- Criterion, fixed before the results were examined: the direct-question pattern is confirmed for a measure if the host-gender × guest-gender interaction of the within-host test has the same sign (guest-gender effect positive for male hosts, negative for female hosts) and $p < 0.05$.
- The within-host test and its interaction are exactly those of Section 6 (percentile ranks within host, 10,000 within-host permutations of guest gender). The guest-gender effect is also tested separately within male hosts and within female hosts, and the interaction is re-run leaving out one host at a time (3,000 permutations each).
- Agreement between the two ways of measuring host questions: Spearman correlation between the direct-question rate and each “ask” measure across episodes, and within host (on within-host percentile ranks).

8.4 How It Was Run

```
sbatch sbatch/run_dialogue_classification.sbatch          # first 200 episodes
```

```
sbatch sbatch/run_dialogue_classification_remaining.sbatch # remaining 218
```

```
python src/combine_dialogue_acts.py
```

```
python src/stat_analysis/ask_within_host_check_v2.py
```

The classification runs on a GPU in the whisperx_cudnn8 environment; the other two scripts run in the convokit environment.

8.5 Output Files

File	Contents
results/dialogue_acts/all_418/ dialogue_act_labels.csv	Label of every segment (68 MB; not in git)
results/dialogue_acts/all_418/ dialogue_act_rates_by_episode_role.csv	Share of each label per episode and role
results/stat_analysis/ ask_within_host_check_v2_episodes.csv	Host question measures per episode
results/stat_analysis/ ask_within_host_check_v2_agreement.csv	Agreement between measures
results/stat_analysis/ ask_within_host_check_v2_within_host.csv	Within-host test and interaction
results/stat_analysis/ ask_within_host_check_v2_by_host_gender.csv	Guest-gender effect within male / female hosts
results/stat_analysis/ ask_within_host_check_v2_per_host.csv	Guest-gender effect per host
results/stat_analysis/	Interaction leaving out one host

ask_within_host_check_v2_leave_one_host_out.csv	
results/stat_analysis/ ask_within_host_check_v2_by_dyad.csv	Medians by dyad

8.6 Results

Agreement between the direct-question rate and the “ask” measures (Spearman correlation):

Measure	Across episodes	Within host
“ask” per 1k host words	0.74	0.77
“ask” share of host segments	0.63	0.66

Both are strongly related to the direct-question rate, also within a host's own episodes: the two methods largely count the same host questions.

Guest-gender effect with the host held fixed (percentile rank of female-guest minus male-guest episodes), separately for male and female hosts, and the difference between them (the interaction):

Measure	Male hosts (4 hosts, 270 eps.)	Female hosts (4 hosts, 104 eps.)	Interaction	Meets criterion
Direct questions per 1k words (Section 1)	+0.145 (p 0.0002)	-0.205 (p 0.0003)	+0.350 (p <0.0001)	Yes
“ask” per 1k host words	+0.093 (p 0.0244)	-0.187 (p 0.0012)	+0.280 (p 0.0003)	Yes
“ask” share of host segments	+0.054 (p 0.1874)	-0.100 (p 0.0865)	+0.154 (p 0.0311)	Yes

The direct-question row repeats Section 6 for comparison. Both “ask” measures meet the criterion. For “ask” per 1,000 words the guest-gender effect is significant within both male and female hosts; for the segment share both halves point the same way but neither is significant on its own.

Guest-gender effect per host (percentile rank, female minus male guests):

Host	Sex	Direct questions	“ask” per 1k words	“ask” share
Faye Dsouza	F	-0.19	-0.19	-0.19
Karishma Mehta	F	-0.21	-0.16	-0.01
Pooja Dhingra	F	-0.40	-0.30	-0.30
Soha Ali Khan	F	+0.50	-0.50	-0.50
Prakhar Gupta	M	-0.04	-0.03	-0.20
Raj Shamani	M	+0.14	+0.05	-0.02

Ranveer A	M	+0.18	+0.21	+0.22
Samdish Bhatia	M	+0.24	-0.01	+0.08

All three female hosts with several episodes of each guest gender ask female guests fewer questions on all three measures (Soha Ali Khan, with one episode of each, is too few to read). Among male hosts, Ranveer A asks female guests clearly more questions on every measure; for Raj Shamani, Prakhar Gupta and Samdish Bhatia the “ask” effects range from -0.20 to +0.08 and do not show the pattern on their own.

Leaving out one host at a time, the interaction stays significant for “ask” per 1,000 words (p at most 0.004) and for direct questions (p at most 0.001). For the segment share it is significant in 6 of 9 cases and not when Ranveer A (p 0.37), Faye Dsouza (p 0.17) or Pooja Dhingra (p 0.07) is left out.

8.7 Interpretation: What These Results Suggest

- **The cross-gender question pattern is confirmed by the dialogue-act label.** The pattern found with direct questions in Section 6 appears again with an independent measurement of questions: male hosts ask female guests more questions than male guests, and female hosts ask female guests fewer. On the comparable measure (per 1,000 words) it meets the pre-set criterion, holds within both host genders, and survives leaving out any single host.
- **The female-host side is the more robust half.** The female-host effect is larger and consistent across the female hosts. The male-host effect is smaller and rests mainly on Ranveer A, so “female hosts ask female guests fewer questions” is the better-supported half.
- **The segment-based measure is weaker.** The share of segments is noisier than a rate per word, probably because how WhisperX splits speech into segments varies between episodes; its weaker result is consistent with more noise rather than a different pattern.

8.8 Notes & Limitations

- The check uses the same 418 episodes in which the pattern was first seen. It confirms that the pattern is not an artifact of the word-list measure, but it is not a replication on new data; the result should still be presented as exploratory and confirmed by a second measure.
- The dialogue-act model was trained on English conversation corpora and was not audited on these podcasts, which mix English and Hindi; its errors on code-mixed speech are unknown.
- Segments are classified with only the previous segment as context, and they include the first and last minute of each episode, unlike Sections 1-7.
- Question counts do not distinguish types of questions (e.g. personal vs. professional), which could differ by guest gender even where the number of questions does not.